MCV4U1 Name: ____________________________ Period: _____

ASSESSMENT FOR LEARNING: OPTIMIZATION

1. A particle moves according to the equation $s(t)=\frac{5-t}{t+1}, t \geq 0$ where ***t*** is in seconds and ***s*(t)** is the distance in meters. Determine the following:
   a) the velocity of a particle when its acceleration is 1.5 m/s^2.**[4]**

   b) Was the particle speeding up or slowing down at t = 3s? **[3]**

2. A 1500m^3 rectangular water storage tank is to be built such that the length of tank is double the height of the tank. If the base and top of the tank cost \$350/m^2 to fabricate and the sides cost \$275/m^2 to fabricate, determine the minimum cost of the tank. **[5]**

3. A rectangular field, 3200 m^2 in area, is to be fenced off along the straight bank of a river, no fence being needed along the bank. What dimensions should the field have, if the amount of fencing is to be a minimum? **[4]**

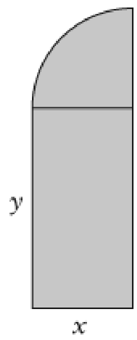
4. A sailor in a boat 4 km off a straight coastline wants to reach a point on shore 10 km from the point directly opposite her present position in the shortest possible time. She can row at 4 km/h and walk at 5 km/h. Toward what point on the shore should she steer? **[5]**

5. A rectangle lies in the first quadrant, with one vertex at the origin and two of the sides along the coordinate axes. If the fourth vertex lies on the line defined by $x + 2y - 10 = 0$ find the dimensions of the rectangle with the maximum area. **[4]**

6. A cylinder is inscribed within a sphere with radius *R* metres. Determine the dimensions of the largest possible cylinder that can be inscribed within the sphere. **[4]**

7. At noon a car is driving west at 55 km/h. At the same time, 15 km due north another car is driving south at 85 km/h. At what time are they closest and what is the closest distance for the planes? **[5]**

8. A unique window has the shape of a rectangle with a quarter circle on top. What height, width, and radius will maximize the enclosed area if the outside perimeter of the entire window is 4 m? **[5]**



Thinking and Inquiry

1. Triangle ABC is isosceles with AB = AC = 5 cm and BC = 6 cm. Another isosceles triangle is formed with one vertex at M, the midpoint of BC, and the other vertices P and Q being any two points on AB and AC respectively such that MP =MQ. Find the greatest area of the triangle MPQ. **[5]**